

MAYANK YADAV

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EDUCATION

Delhi Technological University	May 2027
<i>B. Tech. Computer Science and Engineering</i>	CGPA: 8.99
Central Board of Secondary Education (Class 12th)	May 2023
<i>Computer Science, Mathematics, Physics, Chemistry, English, Physical Education</i>	Grade: 97.4%
Central Board of Secondary Education (Class 10th)	August 2021
<i>English, Hindi, Mathematics, Science, Social Science, Sanskrit</i>	Grade: 97.8%

TECHNICAL SKILLS

Languages: C/C++, Python, Java, HTML, Tailwind-CSS, JavaScript, TypeScript, SQL, $\text{\LaTeX}$
Tools: Git/GitHub, Docker, Kubernetes, Harness CI/CD, Google Cloud Platform, Linux, VS Code
Frameworks: React, Node.js, Flask, FastAPI
Libraries: PyTorch, Detectron2, TensorFlow, Keras, pandas, NumPy, Matplotlib, NLTK, Scikit-Learn, OpenCV, LangGraph

EXPERIENCE

- Open Source Developer, Summer of Bitcoin (SeedSigner)** May 2026 – August 2026
- Developed a stateless secure bootloader for *ESP32-P4 (RISC-V)* and *ESP32-S3 (Xtensa)* to verify and execute MicroPython directly from SD into PSRAM via Cache MMU remapping, preventing secret persistence in flash.
 - Resolved critical FreeRTOS early-boot deadlocks and cache coherency faults by safely intercepting ESP-IDF hardware resets and manually aligning `.data` and `.bss` memory segments during the MMU handoff.
 - Devised a physical anti-phishing defense that fills unallocated flash with TRNG data to compute a 4-word BIP-39 visual proof on boot, validating tamper detection against 16 hardware attack vectors (TOCTOU, bit-flips, evil-maid swaps).
- Technology Program Intern, Wells Fargo, Hyderabad** May 2026 – July 2026
- Built a full-stack agentic intake workflow using React, FastAPI, and LangGraph to automate L3 data application triage, cutting initial Jira screening from 10+ days to near-instant validation.
 - Designed an 11-stage autonomous pipeline featuring an upfront validation layer to automate the end-to-end request lifecycle across mapping, design, QA, and peer review checkpoints.
 - Automated generation of Data Warehouse Design workbooks and QA SQL packs by integrating with an existing Informatica generator to produce downstream ETL artifacts.
 - Orchestrated the application's release on Kubernetes using GitHub Actions and Harness CI/CD, and added a role-based admin dashboard with Server-Sent Events (SSE) for real-time tracking.
- Web Developer Intern, DTU, Delhi** October 2025 – March 2026
- Co-Developed a centralized IQAC data management suite using the MERN stack, deployed across 15+ departments and used by 500+ faculty members for structured academic and administrative data collection with aggregated dashboards.
 - Engineered a 3-tier APAR workflow system (Faculty → Reporting Officer → Reviewing Officer) with automated data pre-filling from the IQAC DB, ensuring data consistency and reducing manual redundancy.
 - Implemented JWT-based authentication, role-based access control (RBAC), and WebSockets (Socket.io) to enable secure access control and real-time update propagation across hierarchical user roles.
- Research Intern, MLR, DTU, Delhi** June 2025 – December 2025
- Developed 5+ deep learning models for vision-based pose estimation of non-cooperative satellites using PyTorch framework.
 - Implemented and optimised multiple SOTA deep learning architectures including SPN, PVNet, ViT Pose, and SWIN Transformer for robust 6-DoF satellite pose estimation; applied RANSAC and Perspective-n-Point solver.
 - Designed a novel SWIN-based pixel-voting mechanism that integrates transformer attention with 4-layer decoder for PVNet's voting strategy.
 - Conducted experiments on SPEED dataset containing synthetic spacecraft images and SPEED+ dataset with hardware-in-the-loop test images.

PROJECTS

- Sherlock** | [GitHub Link](#) | [Deployed Link](#) | *TypeScript, Next.js, React, Tailwind CSS, Vitest* March 2026
- Built a local-first Bitcoin chain analysis tool that parses raw Bitcoin Core .dat files (blk, rev, xor) to reconstruct transaction graphs without full node indexing or external API dependencies.
 - Implemented 10+ rule-based on-chain heuristics including CoinJoin fingerprinting, UTXO consolidation, peeling chain detection, batch payments, and OP_RETURN extraction for transaction pattern classification.
 - Engineered an interactive Next.js 15 dashboard with TypeScript and Tailwind CSS v4 for visualizing block data, transaction flows, and script distributions, with Vitest test coverage and deployed on Vercel.
- GraphGrad** | [GitHub Link](#) | [Deployed Link](#) | *TypeScript, Next.js, React, Tailwind CSS, React Flow, Vitest* March 2026
- Built an interactive computation graph visualizer for learning forward passes and backpropagation, parsing mathematical equations into React Flow graphs with real-time gradient flow visualization.
 - Implemented a custom autodiff engine supporting 13 operations with forward evaluation and backward gradient propagation, inspired by Karpathy's micrograd.
 - Developed with Next.js 16, React 19, TypeScript, and Tailwind CSS v4, featuring light/dark mode, equation-to-graph generation via mathjs, and a Vitest test suite.
- Vulcan-16** | [GitHub Link](#) | [Deployed Link](#) | *HDL, Python, Assembly, Compiler Designing, Jack, Git, React, Flask* January 2025
- Architected a 16-bit Harvard-architecture CPU in HDL, including core logic, memory, and I/O interfaces.
 - Created a 3-phase compilation pipeline and integrated web-based development environment.
 - Designed an OS with 8 modular services for memory management, graphics, I/O, and utilities.
- Mentoring Portal** | [GitHub Link](#) | [Deployed Link](#) | *React, Tailwind, Node.js, Redux Toolkit, Express, MongoDB* December 2024
- Developed a full-stack mentorship portal enabling secure mentor-student engagement with Q&A forums and resource library for 5 competitive exams.
 - Implemented end-to-end features using React, Redux Toolkit, Express, and MongoDB, delivering 4+ core modules.
 - Deployed and maintained a scalable platform serving 50+ underprivileged students for career guidance.
- Int-O-View** | [GitHub Link](#) | [Deployed Link](#) | *Python, TypeScript, React, Express, Flask, MongoDB, LangChain* August 2024
- Engineered an AI-Agent based interviewer app using Gemma 2, Qwen QwQ, and ElevenLabs for realistic voice synthesis.
 - Integrated Supabase embeddings and LangGraph StateGraph for context retrieval powering 100 real-time interview simulations and LLM-driven question generation.
 - Collaborated in a 5-member team to secure 1st place in the SIH internal round, surpassing 200+ competing teams.
- Lung Cancer Detection** | [GitHub Link](#) | *Python, PyTorch, Detectron2, Pydicom* March 2025
- Transformed the LIDC-IDRI dataset by converting 3D DICOM images into 2D JPEG format using advanced windowing techniques, and reformatted XML annotations into the COCO JSON standard.
 - Developed a two-stage lung nodule detection pipeline having a Cascaded Mask R-CNN (trained on VESSEL12 dataset) for lung segmentation, and a Faster R-CNN for nodules localisation and prediction of 5 key radiological features—malignancy, margin, sphericity, texture, and spiculation—to aid in diagnostic decision-making.
 - Achieved segmentation accuracy of 81.98% and bounding box accuracy of 93.62% on the test set in the segmentation model, and Validation set bounding box AP50 scores of 24.97 (malignancy), 24.34 (sphericity), 23.16 (margin), 23.26 (spiculation), and 16.76 (texture).
- Summer ML Projects** | [Github Link](#) | *Python, PyTorch, Transformers* June 2024
- Developed a collection of 7 machine learning projects during the Summer School on AI by MLR, DTU, demonstrating core concepts in ML and deep learning.
 - Implemented a Generative Pre-trained Transformer 2 (GPT-2) from scratch for text generation and a Vision Transformer (ViT) for image classification.
 - Built 3 different image captioning models implementing "Where to put the Image in an Image Caption Generator" paper using a combination of CNN and RNN/LSTM architectures and trained on Flickr 8k images.
- Facial Expression Recogniser** | [GitHub Link](#) | [Kaggle Link](#) | *Python, TensorFlow, Keras, OpenCV, SMOTE* May 2024
- Built a real-time facial emotion recognition system on the FER2013 dataset, addressing severe class imbalance using SMOTE, random oversampling, and data augmentation to improve model generalization.
 - Developed and evaluated 50+ distinct neural network architectures from scratch, spanning fully connected networks to convolutional networks, systematically comparing depth, layer size, and activation configurations.
 - Explored transfer learning with pre-trained Inception and VGG16 models, achieving competitive accuracy through fine-tuning, with a public Kaggle notebook documenting the full experimentation pipeline.

OPEN SOURCE CONTRIBUTIONS

Shopstr TypeScript, Next.js	April 2026
- Contributed 7+ merged bug-fix and feature PRs to a global permissionless Nostr marketplace for Bitcoin commerce (100+ stars), covering server-side pagination with advanced database filtering, profile image upload, and marketplace listing routing. - Fixed client-side issues including navbar profile flickering and deduplication of repeated profile-fetch requests, improving UI stability and load performance.	
BitcoinFuzz C++, Python, Docker	May 2026
Merged 2 PRs to a differential fuzzing project (84+ stars) that compares Bitcoin/Lightning protocol implementations, fixing the Docker build by updating auto_build.py to initialize the secp256k1 submodule for custom-mutator targets and correctly skipping empty-input/empty-output PSBT matches during fuzzing.	
SeedSigner Emulator Python	April 2026
Fixed the MockST7789 initialization TypeError by correcting the display library's constructor call, resolving a blocking startup failure in the emulator.	
NN-SVG GitHub PR Link Deployed Link JavaScript, Three.js	July 2025
Delivered a key feature to NN-SVG (5.6k+ GitHub stars), a tool used by ML researchers to generate publication-ready neural network diagrams, enabling users to upload and render their own images as the input layer texture.	

ACHIEVEMENTS

NTSE Scholar: Recognized for outstanding performance in the NCERT's national-level scholarship examination, placing among the top 1000 scholars.
Reliance Foundation Undergraduate Scholarship: Selected as one of 5,000 nationwide for academic support and leadership development.
CBSE Certificate of Merit in Mathematics: Achieved 100 in Mathematics, ranking in the top 0.1 % in the 2023 All India Secondary School Examination.

POSITION OF RESPONSIBILITY

AlgoRave Co-Head	September 2023 - Present
- Authored and validated problems for 3+ contests on CodeForces, utilizing Polygon platform for problem development and testing. - Drafted 2+ detailed editorials for the contest problems, including solution approaches, algorithms, and code implementations. - Organised contests that attracted 100+ participants, enhancing engagement and competitive programming culture within the university	

LICENSES & CERTIFICATIONS

Machine Learning Specialization Stanford Online / DeepLearning.AI	Dec 2023
Supervised Machine Learning, Advanced Learning Algorithms, Unsupervised Learning & Recommenders	
Foundations of Cybersecurity Coursera / Google	Oct 2025
Hugging Face Agent Course Hugging Face	May 2025
Fundamentals of MCP Hugging Face	May 2025
Introduction to Deep Learning Kaggle	Jan 2024
Getting Started with AI on Jetson Nano NVIDIA	Apr 2025
Google Cloud Computing Foundations (5 courses), Prompt Engineering & GenAI Google	Oct 2023
The Joy of Computing using Python NPTEL	Nov 2025
Human Computer Interaction NPTEL	May 2026